

VENKATA KALYANAM

Texas | +1 (346) 291-0976 | venkatak1611@gmail.com | [LinkedIn](#) | [GitHub](#)

SUMMARY

Full Stack Developer with **4+ years of experience** developing **scalable enterprise applications** using **Java Spring Boot, React.js, AWS cloud services, and CI/CD automation**. Skilled in designing **RESTful APIs, event-driven microservices, and DevOps pipelines** across **finance, asset management, and audit domains**. Experienced in optimizing backend systems with **PostgreSQL, MongoDB, Redis caching**, and containerized deployments using **Docker, Jenkins, and Kubernetes**, delivering solutions supporting **12+ cross-functional teams** and processing **4,200+ daily transactions**.

EXPERIENCE

University of Houston | Houston, TX USA

May 2025 – Present

Software Developer Intern

- Built an automated evaluation platform using **ReactJS, Flask/Node.js, Jenkins, and Docker**, streamlining deployment for **50+ student projects** and cutting faculty grading time from **15 to 3 hours weekly**.
- Instructed **AWS Academy Cloud Operations labs**, teaching frontend-backend integration, **cloud deployment (EC2, S3, CloudFront)**, and performance monitoring, with **180+ students** completing labs in **2 weeks**.
- Maintained and enhanced university websites built on **ReactJS and Node.js**, migrating from **manual deployments to GitHub Actions–based CI/CD pipelines** for consistent, repeatable releases.
- Automated infrastructure provisioning with **Terraform, defining VPC networking, AMIs, Load Balancers, Auto Scaling Groups, and S3 static hosting** to standardize app environments.
- Mentored **200+ students** on **CI/CD workflows** and application optimization, leading to **10 projects** being featured in the university's annual tech showcase.

Dell Technologies | India

February 2021 – July 2023

Full Stack Developer

- Engineered **RESTful APIs** using **Java Spring Boot** and **PostgreSQL** to support enterprise asset management workflows, enabling seamless data access for **12 cross-functional teams** across finance, logistics, and **IT operations**.
- Architected **event-driven microservices** with **AWS Lambda, API Gateway, and SQS** to process **4,200+ daily** transaction events, reducing system coupling and improving scalability for distributed teams.
- Collaborated with **UX designers** and product managers to deliver **18 responsive React.js UI components** with **Bootstrap and TypeScript**, improving user satisfaction scores and streamlining **asset tracking interfaces**.
- Automated **CI/CD pipelines** using **Jenkins, Docker, and Terraform** to deploy containerized **Java applications** on **AWS ECS**, reducing deployment time from **45 minutes to 8 minutes** and eliminating manual configuration errors.
- Optimized **backend services** by refactoring **Spring Data JPA queries** and implementing Redis caching, cutting **API response times** from **1.2 seconds to 320 milliseconds** for high-traffic asset retrieval endpoints.
- Mentored **3 junior developers** on **Agile best practices** and code quality standards using **Jira and Git workflows**, fostering a collaborative environment that accelerated sprint velocity by delivering **22 story points per two-week cycle**.

KPMG | India

June 2020 – January 2021

Junior Software Developer

- Engineered a **Spring Boot microservice** for real-time transaction monitoring, integrating **AWS Lambda** for event-driven processing and cutting false-positive alerts from **210 to 42 weekly** across **3,000+ daily transactions**.
- Automated **CI/CD pipelines** using **Jenkins** and **Docker**, reducing deployment failures by 6 incidents per quarter and accelerating release cycles from **5 to 2 days**, earning recognition as the top contributor in the DevOps sprint retrospective.
- Designed a **React.js dashboard** with **Node.js backend APIs** to visualize **audit trails**, cutting manual report generation time from **4 hours to 20 minutes** and adopted by **150+ internal auditors** within **3 months**.
- Refactored **PostgreSQL queries** and optimized **AWS RDS indexing**, dropping database response times from **3.2 to 1.1 seconds**, directly improving the month-end closing process for **10+ client engagements**.

TECHNICAL SKILLS

- **Languages & Programming:** Java, Python, JavaScript (ES6+), TypeScript, Groovy, SQL, HTML5, CSS3
- **Frameworks & Libraries:** Spring Boot, Spring Core, Spring MVC, Spring Data JPA, Spring Security, Hibernate, JPA, Flask, React.js, Angular, Bootstrap, SASS, jQuery, Design Patterns
- **Backend & APIs:** RESTful APIs, JSP, Node.js (secondary), Express.js, RESTful Web Services, Swagger UI, Postman
- **Databases:** PostgreSQL, MongoDB, Oracle SQL, DynamoDB, Cassandra, Aurora, RDS
- **Cloud & DevOps Tools:** AWS (IAM, EC2, S3, ECS, EKS, Lambda, Systems Manager, Load Balancing, AutoScaling, API Gateway, EventBridge, CloudWatch, CloudFormation, VPC, SQS, SNS, RDS, Aurora, DynamoDB, CDK), Azure DevOps, Google Cloud Run, Docker, Terraform, Jenkins, Kubernetes, Git, GitHub, GitHub Actions, Grafana, Linux, AWS CLI
- **Testing & Debugging:** JUnit, Mockito, Postman, Unit Testing, Test Driven Development, Form Validation
- **CI/CD & Deployment:** GitHub Actions, Jenkins, Docker, Azure Pipelines, Maven, Gradle
- **Design & Prototyping:** Figma, Responsive Web Design
- **Development Practices:** Agile (Scrum), Git, Jira, Confluence, Version Control

PROJECTS

Federated Learning for Image Classification:

- Implemented privacy-preserving Federated Learning algorithms (FedAvg, FedDyn, SCAFFOLD) using **ResNet-18** and **ResNet-50** across **3 datasets (CIFAR-10, Fashion-MNIST, Lung X-Ray)**, enabling decentralized training on **10 simulated clients** while maintaining data privacy.
- Evaluated model robustness under **IID** and **Non-IID** distributions by analyzing **accuracy** and **loss metrics**, identifying performance variations of up to **4.3 standard deviation points** across heterogeneous clients.

Expense Lens:

- Developed an **Android expense tracking app** in **Kotlin**, integrating **AWS Textract** for **OCR**, **Flask REST APIs** on **EC2**, **Lambda** for ML classification, and **RDS** for financial data management.
- Architected **cloud storage** with **AWS S3** for active receipts and **Glacier** for archival, processing **500+ receipts** with automated metadata extraction and efficient retrieval workflows.

EDUCATION

Master of Science in Computer Science | Houston, TX

August 2023 – May 2025

University of Houston | GPA: 3.8/4.0

CERTIFICATIONS

- [AWS Certified Solutions Architect – Associate](#)
- [AWS Certified Cloud Practitioner](#)
- [Java for Android – Vanderbilt University](#)
- [Kotlin for Java Developers – JetBrains](#)